

Course Code : ECT 354

GVHW/RW – 23 / 1216

**Fifth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

DIGITAL SIGNAL PROCESSING

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
 - (2) Use of Non – programmable calculator is allowed.
 - (3) Assume suitable data wherever necessary.
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1. (a) Compute the circular convolution of two sequences using DFT-IDFT approach
 $\mathbf{x(n) = n - 1 \text{ for } 0 \leq n \leq 3}$ and $\mathbf{h(n) = \cos(n\pi)}$ for $\mathbf{0 \leq n \leq 3}$. 5(CO1,2)
 - (b) Compute 8-Point DFT of sequence $\mathbf{x(n) = \{1, 1, 1, 1, 1, 1\}}$. Use Radix-2,
DIF-FFT Algorithm. 5(CO1,2)
 2. (a) Realize the IIR system given by difference equation :
 $\mathbf{y(n) + (1/4) y(n-1) - (1/8) y(n-2) = x(n) - 2x(n-1) + x(n-2)}$ in :
(1) Cascade form.
(2) Parallel form. 6(CO3)
 - (b) Realize the given system function using minimum number of multipliers :
 $\mathbf{H(z) = 1 + 0.5z^{-1} + 0.25z^{-2} + 0.75z^{-3} + 0.25z^{-4} + 0.5z^{-5} + z^{-6}}$. 4(CO3)
 3. Design and realize digital IIR Butterworth LPF filter using bilinear transformation with
following constraints. Assume $\mathbf{T = 1 \text{ sec}}$:
 $\mathbf{0.9 \leq |H(e^{jw})| \leq 1 \text{ for } 0 \leq w \leq \pi/2}$
 $\mathbf{|H(e^{jw})| \leq 0.2 \text{ for } 3\pi/4 \leq w \leq \pi}$ 10(CO3)

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4. (a) A low pass filter is to be designed with the following desired frequency response :
- $$H_d(e^{j\omega}) = e^{-j2\omega}, \quad -\pi/4 \leq \omega \leq \pi/4$$
- $$0, \quad \pi/4 \leq \omega \leq \pi$$
- Determine the filter coefficients $h_d(n)$ if the window function is defined as
- $$w(n) = 1, \quad 0 \leq n \leq 4$$
- $$0, \quad \text{otherwise}$$
- Also, determine the frequency response $H(e^{j\omega})$ of the desired filter. 8(CO3)
- (b) Distinguish between FIR and IIR filters. 2(CO1)
5. (a) For the signal given by
- $$x(n) = a^n, \quad n \geq 0 \quad \text{for } |a| < 1$$
- $$= 0, \quad \text{otherwise}$$
- (1) Find the expression and plot decimated signal with decimation by factor $D = 3$.
- (2) Find the expression and plot Interpolated signal with Interpolation factor $I = 3$. 6(CO4)
- (b) Write a short note on subband coding of speech signal. 4(CO1,4)
6. (a) Explain with functional block diagram the architecture of DSP Processor. 6(CO1,5)
- (b) State the advantages and applications of DSP processors. 4(CO1,5)

